

JBIG Standard

ASPECT	DETAILS
Full Form	Joint Bi-level Image Experts Group
Standard	ISO/IEC 11544 (1993) ISO → International Organization for Standardization IEC → International Electrotechnical Commission
Compression Algorithm	Context-based Arithmetic Coding (instead of LZW like GIF)
Compression Nature	Lossless (perfect reconstruction of black/white pixels)
Image Type Supported	Bi-level (1-bit per pixel) images → each pixel is black (0) or white (1)
Key Idea	Predict the probability of the next pixel using the values of surrounding (neighbor) pixels → compress using arithmetic coding
Working Process (Pipeline)	1. Input = scanned bi-level image 2. Pixels grouped into a context window (e.g., 10 nearby pixels) 3. Based on this context, assign probability to the current pixel 4. Apply arithmetic coding to encode pixel sequence efficiently 5. Store compressed data in JBIG format
Compression Efficiency	Very high for text, faxes, and scanned documents (2 to 5 times smaller than fax Group 4)
Data Structure	JBIG file contains: Header + Arithmetic-coded data + optional adaptive templates for context
Transparency Support	Not applicable (only 1-bit images, no colors)
Animation Support	Not supported (only still documents/images)
Interlacing/Progressive	Yes → supports progressive transmission (image appears gradually as more data arrives)
Strengths	- Better compression than run-length encoding (Fax G3/G4) - Standardized & widely adopted in document imaging - Lossless, so text/images retain accuracy - Supports progressive transmission. Group 3 (G3): Run-length encoding (basic compression).

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	Group 4 (G4): Two-dimensional run-length coding (improved, widely used).
Limitations	- Works only for bi-level images (no grayscale or color) - Complex arithmetic coding increases encoding time - Largely replaced by JBIG2 in modern systems
Applications	- Fax transmission (ITU-T T.82 standard) - Scanned documents storage - OCR preprocessing - Digital libraries & archival. International Telecommunication Union – Telecommunication Standardization Sector
Successor	JBIG2 (ISO/IEC 14492, 2000) → supports higher compression, pattern matching for repeated shapes (e.g., letters in text)
Comparison with GIF	- GIF uses LZW + indexed colors (up to 256 colors) → good for web/animations - JBIG uses Arithmetic Coding + 1-bit pixels → good for text/fax documents